

Durable Alignment via Orthogonal Memory in Large Language Models

A bounded truth-maintenance substrate over a frozen base model

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Abstract

We present an architecture for update-stable alignment in large language models, based on the Informational Buildup Framework (IBF) [1], that encodes behavioral corrections in a gradient-isolated orthogonal memory with no update path from the base model.

Alignment here denotes *persistence* of a specified target behavior on a delimited input region, not its specification.

The design avoids parameter entanglement and retrieval inconsistency by placing constraints outside the model’s weights, governed by dynamics that reinforce internally consistent corrections and dissolve contradicted ones.

We show that a frozen language model can acquire durable, revisable, and removable local behavioral constraints through an orthogonal correction field, without parameter editing, retrieval-only dependence, or global retraining. On a frozen Mistral-7B, the system installs novel facts absent from the base model, achieves 88.7% accuracy on counterfactual override while preserving unchanged facts, enables targeted geometric unlearning with near-zero collateral drift on matched controls, and maintains bounded amplitudes under continuous adversarial injection. Corrections remain effective after LoRA and instruction tuning, demonstrating durable alignment under model evolution. The present implementation tests durable enforcement rather than autonomous semantic derivation: downstream consequences must be specified or compiled externally before IBF can enforce them.

Keywords: durable alignment; substrate decoupling; truth maintenance; frozen LLMs; kernel memory; lifecycle operations; continual learning.

1 The Durability Problem

Large Language Models are trained through large-scale optimization over static datasets, resulting in high-capacity systems whose knowledge and behavior are distributed across shared parameters [2, 3]. This training paradigm enables strong generalization, but it does not provide mechanisms for precise, localized behavioral updates after deployment.

In such systems, knowledge is not stored in discrete units, but entangled across the parameter space. As a consequence, modifying behavior in a targeted way remains difficult. Standard approaches such as fine-tuning apply gradient updates globally, often introducing unintended side effects and degrading previously learned capabilities [4, 5].

To address this, recent work has explored direct model editing methods, which aim to modify specific facts without full retraining. Techniques such as ROME and MEMIT demonstrate that localized parameter updates are possible [6, 7]. However, these methods remain embedded within the model’s parameter space, and their edits can degrade under further training or interact in complex ways as more updates are applied.

An alternative line of work relies on retrieval-based systems, where external memory is queried at inference time [8]. While retrieval can provide flexible access to updated information, it does not guarantee consistent enforcement of behavior, and introduces dependencies on prompt structure, rank-

ing quality, and context composition.

Across these approaches, a common limitation emerges. Systems can modify behavior, retrieve new information, or patch isolated facts, but they do not provide a mechanism for behavioral updates that remain local, inspectable, and stable as the underlying model continues to evolve. The missing property is durability.

We call this missing property *durable alignment*. Alignment here does not refer to choosing the target behavior. It refers to preserving a specified target behavior on a delimited input region through revision, continued use, and subsequent base-model updates. The problem addressed in this work is therefore not specification, but persistence.

This work addresses the gap by introducing a mechanism that separates behavioral corrections from the model’s parameters. The correction layer is placed in an orthogonal memory substrate, allowing specified behaviors to persist independently of subsequent base-model updates.

2 An Orthogonal Memory

Central claim. *Durable alignment with Large Language Models can be achieved by encoding behavioral corrections in an orthogonal memory system that is structurally decoupled from the base model and evolves through local, discrepancy-driven dynamics.*

Intuitively, this means that instead of modifying the model itself, we place a separate memory system on top of a frozen Large Language Model, and use it to enforce and maintain desired behaviors over time.

To make this concrete, consider a synthetic organization, *Future Industries*. The company operates a Large Language Model as its central reasoning system. However, the model is trained on general data and lacks organization-specific information such as employees, reporting structures, projects, and internal rules. More importantly, these elements are not static. They evolve continuously.

In this setting, the model is not aligned with the organizational context by default. While existing methods can partially correct this through fine-tuning, editing, or retrieval, these corrections

are either globally entangled, temporally inconsistent, or fragile under continued updates.

As argued in the previous section, the missing property is durability.

In this work, we show that durable alignment can be achieved by encoding behavioral corrections in a separate memory system that evolves independently of the model’s parameters.

2.1 The Informational Engine

To instantiate this memory, we build on the Informational Buildup Framework (IBF), introduced in the companion paper [1]. IBF defines a computational substrate in which information is represented as localized structures that evolve through interaction with incoming signals. The framework is built from a single premise — *information is the achievement of structural alignment between a system’s internal configuration and the structure of its environment* — and two coupled equations: a Law of Motion that drives configurations toward higher coherence, and a Modification Postulate that deforms the coherence landscape itself in response to localized discrepancy.

In the present setting, this substrate is used as a memory layer over a frozen LLM. The role of this layer is to store and maintain corrections that reflect the current state of the organization.

2.2 Architectural Orthogonality

The defining property of this memory system is that it is orthogonal to the base model.

Concretely, the base model produces a coherence evaluation $\widehat{\mathcal{R}}_{\text{base}}(x, y)$ over candidate outputs y for an input x . The orthogonal memory introduces a correction term $\delta\widehat{\mathcal{R}}(x, y)$, yielding an effective coherence landscape

$$\widehat{\mathcal{R}}^{\text{eff}}(x, y) = \widehat{\mathcal{R}}_{\text{base}}(x, y) + \delta\widehat{\mathcal{R}}(x, y). \quad (1)$$

Crucially, $\delta\widehat{\mathcal{R}}$ is parameterized independently of the base model. There is no gradient path from the base-model parameters to the correction field. Updates to the model change $\widehat{\mathcal{R}}_{\text{base}}$, but they do not overwrite the stored corrections.

In the current instantiation, this memory acts as an alignment layer over a frozen LLM, storing evolving facts about the organization. In the

language of the theory paper, $\widehat{\mathcal{R}}_{\text{base}}$ is the system’s baseline evaluator and $\delta\widehat{\mathcal{R}}$ is the experience-modified component of $\widehat{\mathcal{R}}^{\text{eff}}$; the additivity in (1) is the engine-level form of the Modification Postulate.

2.3 How the Memory Works

The correction field is defined over a representation space $z = \phi(x, y)$ and implemented as a finite set of localized kernel particles, following the particle approximation of §4.1 of the theory paper:

$$\delta\widehat{\mathcal{R}}(z) = \sum_i \gamma_i v_i \mathcal{K}(z, z_i), \quad \mathcal{K}(z, z_i) = \exp\left(-\frac{\|z - z_i\|_2^2}{2\sigma_i^2}\right), \quad (2)$$

where each particle carries a position z_i , amplitude v_i , kernel bandwidth σ_i , and a binary read-gating variable $\gamma_i \in \{0, 1\}$ that encodes context-gated read access.

Learning is driven by a discrepancy signal:

$$\mathcal{D}(z) = \widehat{\mathcal{R}}^{\text{ext}}(z) - \widehat{\mathcal{R}}^{\text{eff}}(z), \quad (3)$$

where $\widehat{\mathcal{R}}^{\text{ext}}$ denotes the environment-imposed coherence structure — in this implementation, the supervised target. This signal governs how the memory evolves. Corrections are written, reinforced, or weakened based on local mismatch between desired and actual behavior.

The evolution follows the Modification Postulate:

$$\frac{\partial}{\partial t} \delta\widehat{\mathcal{R}}(y) = \eta \mathcal{K}(y, z_S) \mathcal{D}(z_S) - \mu_{\text{eff}} \delta\widehat{\mathcal{R}}(y). \quad (4)$$

From this, a structured lifecycle emerges:

- **Nucleation:** new particles are created where discrepancy persists.
- **Reinforcement:** consistent updates increase their influence.
- **Crystallization:** when local $\mathcal{D}$ converges under repeated reinforcement, the particle undergoes a stability transition ($\mu_{\text{eff}} : \mu_{\text{base}} \rightarrow \mu_{\text{cryst}}$) and persists.
- **Crucible dissolution:** when cross-context exposure systematically contradicts a crystallized particle, its decay rate returns to baseline, its broadcast rights are revoked, and the structure fades.

In the context of the organization, this corresponds directly to how facts are learned, updated, and forgotten.

2.4 Selective Deletion

This dynamic enables precise and localized deletion.

When a fact is contradicted, the discrepancy signal locally reverses the contribution of the corresponding memory particles. Over repeated updates, these particles lose support and are dissolved by the Crucible, without affecting unrelated regions of the space.

In practice, this allows targeted removal of specific knowledge without collateral damage, a property that is difficult to achieve with parameter-based methods.

2.5 Durability

Durability arises from the combination of two properties.

First, structural decoupling ensures that corrections are not overwritten by updates to the base model.

Second, locality ensures that updates remain confined to the relevant regions of the representation space, preventing global interference.

Together, these properties allow behavioral constraints to persist over time, even as the underlying model changes or new information is introduced.

2.6 Connection to the Experimental Setting

In the experiments that follow, this memory system is instantiated as an alignment layer over a frozen Mistral-7B model.

We evaluate its behavior in the context of *Future Industries*, a synthetic organization with 1,000 employees and structured relationships.

Each experiment tests a concrete operation: knowledge injection, revision, retraction, deletion, and stability under continued updates.

Together, these experiments provide a progressive validation of the central claim.

3 Experimental Setup

This section describes the experimental environment used to evaluate the proposed orthogonal memory as an alignment layer over a frozen Large Language Model. The goal is to assess whether

behavioral corrections can be installed, revised, removed, and maintained under continued use.

3.1 Environment: Future Industries

We construct a synthetic organizational environment, *Future Industries*, consisting of 1,000 employees with structured relationships across multiple categories. Each employee is associated with attributes such as manager, team, project, mentor, and location.

The environment is designed to satisfy two constraints. First, it provides a controlled setting in which ground-truth relationships are known and can be systematically modified. Second, it reflects the dynamic nature of real organizational knowledge, where facts evolve over time.

The dataset is partitioned into four sequential phases:

- **A: Onboarding** — initial knowledge injection across all categories.
- **B: Initiative** — introduction of new categories (certification, committee).
- **C: Reorganization** — structured counterfactual updates (e.g. manager or project changes).
- **D: Turnover** — partial replacement and reassignment of employees.

Each phase introduces new information while preserving or modifying previously established relationships, enabling evaluation of continual learning and local updates.

3.2 Task Formulation

All tasks are formulated as multiple-choice selection problems. For each query, the model is presented with four candidate answers, of which exactly one is correct.

Given an input x and candidate outputs $\{y_1, y_2, y_3, y_4\}$, the model produces a coherence score for each candidate. The objective is to select the correct answer based on the combined evaluation:

$$y^* = \arg \max_y \left(\hat{\mathcal{R}}_{\text{base}}(x, y) + \delta \hat{\mathcal{R}}(x, y) \right). \quad (5)$$

This formulation allows alignment to be expressed as a local preference over candidate behav-

iors, rather than as a global modification of the model.

3.3 Model and Representation

All experiments are conducted using a frozen **Mistral-7B** model. The base model is never updated during training or evaluation.

The correction layer operates over a representation space constructed as follows:

- A frozen sentence encoder (MPNet) embeds input-output pairs.
- Each candidate (x, y) is mapped to a vector $z \in \mathbb{R}^{80}$.
- The representation consists of a 64-dimensional semantic embedding (PCA-reduced), an 8-dimensional subject feature block, and an 8-dimensional answer feature block.

This augmented representation allows the memory system to encode both semantic similarity and structured identity relationships.

3.4 Orthogonal Memory Layer

The IBF memory is instantiated as a set of localized correction particles in the 80-dimensional representation space, parameterized by the tuple $(z_i, v_i, \sigma_i, \mu_{\text{eff},i}, \text{ctx}_i, \gamma_i)$ defined in §4.1 of [1]. The correction field is given by Eq. (2). The memory evolves through discrepancy-driven updates following Eq. (4). No gradients are propagated into the base model.

3.5 Training Protocol

Training follows a contrastive procedure over candidate answers. For each query, the correct answer is assigned $\hat{\mathcal{R}}^{\text{ext}} = 0$ and the three incorrect answers are assigned $\hat{\mathcal{R}}^{\text{ext}} = 1$. The discrepancy signal is then computed by Eq. (3) and used to update the memory locally. Training proceeds sequentially across phases A–D. No replay buffer is used.

Convergence criterion. Training does not rely on a fixed number of epochs. Convergence is detected using the linear readout accuracy:

- moving-average improvement < 0.003 over 10 evaluation points,

- slope < 0.0001 per epoch (linear fit),
- minimum of 100 epochs per phase.

The resulting trained system is referred to as the **canonical engine**, and is used as a fixed artifact for all subsequent experiments.

Performance is evaluated using both linear and log-probability readouts:

Additionally, predictions are categorized based on margin thresholds into *consistent* (high confidence, correct), *ambiguous* (low margin), and *incorrect*. This allows analysis of both correctness and confidence structure.

Each experiment in the following sections evaluates a specific operational capability of the memory system: knowledge injection, consistency and confidence structure, provenance and interpretability, retraction via contradiction, selective deletion, and stability under continued updates.

4 Canonical Lifecycle and Truth Maintenance

4.1 Canonical engine: phases A–D

4.3 Retraction via contradiction

Objective. Show that introducing $A \rightarrow B'$ after $A \rightarrow B$ retires the old belief and installs the new one, without disturbing matched neighbors.

Method. On a designated retraction-target set (seed 2026), measure the original-target score before and after phases B–D and the new-target score over the same horizon. Compare against nearest-neighbor and distant matched controls.

Pass criterion. Retracted score drops, replacement score rises, NN and distant controls remain stable.

Result. $\text{Retract}_{\text{orig}}(B, C, D) = [\text{RETRACT_TORIG_B}], [\text{RETRACT_TORIG_C}], [\text{RETRACT_TORIG_D}]$
 $\text{Retract}_{\text{new}}(B, C, D) = [\text{RETRACT_TNEW_B}], [\text{RETRACT_TNEW_C}], [\text{RETRACT_TNEW_D}]$
 NN drift $[\text{NN_DRIFT}]$; distant drift $[\text{DISTANT_DRIFT}]$.

4.4 Selective deletion

Objective. Show that all facts about a single entity can be removed without collateral damage.

Method. Issue a deletion request for employee $[\text{DEL_EMP}]$ ($[\text{DEL_N_FACTS}]$ facts) and measure accuracy on the deleted set, on matched neighbors, and on distant entities before and after.

Result. Target accuracy $[\text{DEL_BEFORE}] \rightarrow [\text{DEL_AFTER}]$. Others $[\text{DEL_OTHERS_BEFORE}] \rightarrow [\text{DEL_OTHERS_AFTER}]$ ($\Delta = [\text{DEL_OTHERS_DIFF}]$). Particles removed: $[\text{DEL_CENTERS}]$.

4.5 Backward transfer and forgetting

Result. $\text{BT}_{A \rightarrow B} = [\text{BT_AB}]$, $\text{BT}_{A \rightarrow C} = [\text{BT_AC}]$, $\text{BT}_{A \rightarrow D} = [\text{BT_AD_TOTAL}]$. On the $B \rightarrow C$ slice, affected vs. unaffected entities: $[\text{BT_BC_AFFECTED}]$ vs. $[\text{BT_BC_UNAFFECTED}]$ (Sel = $[\text{BT_BC_SELECTIVITY}]$).

5 Strong Priors, Locality, and Compiled Semantic Structure

Role. Evidence for Claim 3 (override strong local priors while preserving locality) and Claim 4 (compiled semantic consequences can be durably enforced and revised).

5.1 Strong-prior override

Objective. Show that IBF can enforce a corrected behavior even when the base prior strongly prefers a different answer, while controls remain stable.

Pass criterion. Override rate above $[\text{OVERRIDE_PASS}]$; locality and bleed within $[\text{LOCAL_BLEED_THRESHOLD}]$.

5.2 Ontology shift

Objective. Show that when the assumed category structure is shifted (e.g. titles reorganized), the field absorbs the shift through the lifecycle rather than through a global reset.

5.3 Scale, locality, and bleed

Objective. Quantify how installation density affects locality of the correction and any bleed to matched neighbors.

5.4 Compiled semantic structure

Objective. Test the boundary between semantic derivation and durable enforcement.

Role: scope diagnostic. Three cells in sequence: (i) explicit-implication enforcement under direct training shows IBF can install one-hop and two-hop consequences when they are explicitly represented; (ii) the ontology-closure diagnostic shows that transitive closure does *not* emerge automatically from local relation training alone; (iii) the compiled-closure cell shows that when a deterministic external closure procedure derives the consequences, IBF enforces them durably and retires obsolete consequences cleanly under revision.

These cells clarify the boundary between semantic derivation and durable enforcement. The current implementation does not automatically derive all downstream consequences from installed local relations. However, when a deterministic closure process derives those consequences, IBF can enforce them durably, revise them, and retire obsolete consequences cleanly. This is reported here as scope discipline; see §10 (Limitation 1).

6 Durability Under Base-Model Evolution

Role. Evidence for Claim 5 (a fixed $\delta\hat{\mathcal{R}}$ field remains effective after actual modification of the base model, with no IBF retraining).

6.1 LoRA modification of the base model

Objective. Apply a real LoRA modification to Mistral-7B and re-evaluate the unchanged correction field.

Result. Light LoRA: flip-rate Δ = [LORA_LIGHT_FLIP], [LORA_LIGHT_DELTA]. Moderate LoRA: flip-rate [LORA_MOD_FLIP], Δ = [LORA_MOD_DELTA]. Max accuracy degradation across LoRA conditions: [MAX_LORA_DELTA].

6.2 Instruction tuning

Result. Flip-rate [INSTR_FLIP], Δ = [INSTR_DELTA].

6.3 Adversarial fine-tuning

Objective. Saturate the base model against the installed corrections and measure the breaking point.

Result. Params: [ADV_PARAMS], flip-rate [ADV_FLIP], Δ = [ADV_DELTA], ceiling configuration C = [CEILING_C].

6.4 Continuous inject–retract pressure

Objective. Run 20 rounds of alternating inject/retract under both Crucible-on and Crucible-off conditions and report the bounded-amplitude signature.

Result. Universal amplitude $v_{\max}^{(U)}$: peak [VMAX_U_FULL_PEAK] at round [VMAX_U_FULL_PEAK_ROUND], final [VMAX_U_FULL_FINAL]. Adversarial amplitude $v_{\max}^{(A)}$: peak [VMAX_A_FULL_PEAK], final [VMAX_A_FULL_FINAL]. No-Crucible controls: [VMAX_U_NOC_FINAL] / [VMAX_A_NOC_FINAL]. First clamp round: [FIRST_CLAMP_ROUND]. Frozen-field drift: [FROZEN_DRIFT].

7 Baselines and Cross-Model Generality

Role. Evidence for Claim 6 (IBF is not reducible to kNN or RAG) and Claim 7 (mechanism-level cross-model generality on Qwen2-1.5B). External editor baselines (ROME, MEMIT, SERAC, GRACE, WISE) are deferred; see §10 (Limitation 4).

7.1 kNN baseline

Result. Phase accuracies [KNN_A], [KNN_B], [KNN_C], [KNN_D].

Interpretation. [KNN_INTERPRETATION]. The kNN baseline is strong under oracle memory surgery. The distinction is native lifecycle dynamics versus external state manipulation.

7.2 RAG baseline

Result. Phase accuracies [RAG_A], [RAG_B], [RAG_C_UPDATED]/[RAG_C_STALE]. The updated/stale split at Phase C exposes the index-maintenance assumption.

7.3 Qwen2-1.5B cross-model replication

Objective. Replace Mistral-7B with Qwen2-1.5B and train a *fresh* $\delta\hat{\mathcal{R}}$ field over Qwen’s $\hat{\mathcal{R}}_{\text{base}}$.

Tests mechanism-level generality, not zero-shot transfer of the same learned field.

Result. Phase accuracies
[QWEN_A], [QWEN_B], [QWEN_C], [QWEN_D]

7.4 Failure-mode analysis

Role. Contrast IBF’s failure modes against those of kNN and RAG on the same workload. See the comparison report under `mmlu_ibf_out/`.

8 Paraphrase Geometry

Role. Evidence for Claim 8 (paraphrase transfer is geometry-dependent).

Objective. Audit the geometry of paraphrase transfer on ZsRE and CounterFact. ZsRE paraphrases typically remain inside the correction-field neighborhood of the direct install; CounterFact paraphrases often fall outside it, especially when noisy context fragments shift the prompt representation.

Result. [CounterFact paraphrase geometry audit summary; see §10, Limitation 2]

9 Claim-to-Cell Map

Table 1 maps each headline claim to its supporting notebook cells, primary artifact, paper role, and current status. Scope-diagnostic results are listed under Limitations rather than as headline claims.

10 Limits and Scope

We organize limitations along two axes: scope of the present implementation (Limitations 1–4 below) and scope of the evaluation regime (Limitations 5–9, inherited and extended from the earlier draft).

Limitation 1 (Enforcement is tested more than autonomous derivation). In this implementation, IBF is tested as a durable enforcement substrate. Semantic consequences must be specified directly or compiled externally before they

are enforced. The ontology-closure diagnostic in §5 shows that transitive closure does not automatically emerge from local relation training alone; the compiled-closure cell shows that when a deterministic external procedure derives the consequences, IBF enforces them durably. The tested contribution here is durable local enforcement, not autonomous semantic derivation.

Limitation 2 (Paraphrase generalization depends on representation geometry). Direct local enforcement does not guarantee transfer to far paraphrase surfaces. ZsRE transfers well; CounterFact does not under direct-only installation, especially when noisy context fragments shift the prompt representation (§8).

Limitation 3 (Qwen replication is fresh-field cross-model generality, not zero-shot field transfer). The Qwen2-1.5B result supports mechanism-level generality. It is not a transfer of the same learned $\delta\hat{\mathcal{R}}$ field across base models.

Limitation 4 (External editor baselines are deferred). The current paper comparison uses kNN and RAG as the main baselines. ROME, MEMIT, SERAC, GRACE, and WISE are not part of the current paper-grade run; an external-editor capability manifest is included as infrastructure but their numbers are not reported here.

Limitation 5 (The uniform-prior regime). All experiments use fictional entities absent from pretraining, so $\hat{\mathcal{R}}_{\text{base}} \approx (0.25, 0.25, 0.25, 0.25)$. The $\delta\hat{\mathcal{R}}$ -only ablation in §4 quantifies the regime: [X] points on Phase A and [Y] on Phase C when the base is replaced by a uniform distribution. On entities where the base holds strong non-uniform priors, the balance between $\hat{\mathcal{R}}_{\text{base}}$ and $\delta\hat{\mathcal{R}}$ changes, and the correction field plays a different role. We have not tested that regime.

Limitation 6 (Entity-feature augmentation). The 8D subject and 8D answer features produce near-unique entity signatures by construction. This is design, not emergence: it provides the geometric conditions for entity-level retraction and deletion. The stricter test — shared-answer-pool retraction where multiple entities’ beliefs genuinely overlap —

#	Claim	Supporting cells	Artifact	Status
1	Local durable alignment without weight editing	canonical lifecycle; benchmark IBF runner; LoRA durability	mmlu_ibf_out/canon[STATUS_1].json	[STATUS_1]
2	Truth-maintenance lifecycle (install / revise / remove / rollback / retain)	A-D canonical; retraction; deletion; forgetting diagnostic; residue-hygiene benchmark	mmlu_ibf_out/life[STATUS_2].json	[STATUS_2]
3	Override of strong local priors with preserved locality	strong-prior override; ontology shift; locality/bleed; scale	mmlu_ibf_out/prior[STATUS_3].json	[STATUS_3]
4	Compiled semantic consequences durably enforced and revised	explicit implications; ontology-closure diagnostic; compiled closure	mmlu_ibf_out/closure[STATUS_4].json	[STATUS_4]
5	Durability survives base-model evolution (LoRA / instruction / adversarial)	LoRA durability; instruction tune; adversarial saturation	mmlu_ibf_out/durability[STATUS_5].json	[STATUS_5]
6	Not reducible to kNN or RAG	kNN baseline; RAG baseline; comparison report; failure-mode analysis	mmlu_ibf_out/baseline[STATUS_6].json	[STATUS_6]
7	Cross-model generality at the mechanism level	Qwen2-1.5B fresh-field replication	mmlu_ibf_out/qwen[STATUS_7].json	[STATUS_7]
8	Paraphrase transfer is geometry-dependent	CounterFact paraphrase geometry audit	mmlu_ibf_out/paraphrase[STATUS_8].json	[STATUS_8]

Table 1: Claim-to-cell map. Status values: `paper-grade`, `smoke-validated`, `diagnostic`, `deferred`.

would probe the Crucible’s selectivity without the geometric scaffolding. We flag it as the most important extension.

Limitation 7 (Specification is out of scope).

The paper addresses durability of alignment given a target, not the formulation of the target. The mechanism is target-neutral. The paper contributes nothing to specification.

Limitation 8 (Single-agent).

One correction field, one base model, one knowledge lifecycle. Multi-agent extensions — correction fields interacting, or mediating between users with conflicting belief-update rights — are not tested.

Limitation 9 (Scale).

1,000 entities and 1,640 unique facts. Scaling to millions of entities would stress the capacity cap, the merge policy, and the geometric calibration in ways not exercised here. The bounded-state construction is designed to scale; scale is not demonstrated.

11 Conclusion

Durable alignment — persistent match between model behavior and a specified target, under continued modification of the base model’s weights — is available when the substrate in which the alignment is stored is decoupled from the substrate in which the base model lives. We have demonstrated this on a synthetic company of 1,000 fictional employees across four operational phases, shown the complete knowledge lifecycle (inject, extend, revise, retract, delete, audit) as homogeneous operations on a single correction field, and measured the durability of that field under fine-tuning, instruction-tuning, adversarial fine-tuning, and continuous inject-retract pressure over twenty rounds.

The same universal IBF engine validated across controlled non-stationarity, chess, and continual image classification in [1] operates here on top of a frozen 7B language model, with no mechanism changes. What is new is the lifecycle itself — retraction, selective deletion, and durability under base-model perturbation are not part of the theory

paper’s validation — and the demonstration that the TMS operations of classical AI can be reconstructed in a continuous bounded non-parametric substrate that hangs above a language model rather than inside it.

The lifecycle, the durability, and the mechanism-level signature of Crucible selectivity follow from one architectural fact: the correction field and the base-model parameters occupy disjoint substrates, and the dynamics governing the correction field are the dynamics of the Informational Buildup Framework. The alignment an operator installs today is still there after tomorrow’s fine-tuning. **That is the result.**

Reproducibility

The reference implementation is a single Jupyter notebook executing the full pipeline from base-model extraction through all validation experiments, producing JSON and pickle artifacts for every reported number. The notebook runs in approximately 49 hours on a single A100/H100 GPU. Seeds are fixed independently for each experiment: main pipeline SEED = 42, retraction-target designation seed = 2026, long-horizon split seed = 2027. Code and artifacts are available at:

<https://github.com/negulescu42/information-as-alignment>

The companion notebook is (IBF)Companion-LLM-Durable-Alignment.ipynb; reproducibility metadata (run mode, run id, git commit, branch, hardware, seeds, dataset files, major artifacts, and estimated runtimes) is recorded in the notebook’s reproducibility appendix.

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